[INCH POUND]
A-A-58034
October 17, 1994

COMMERCIAL ITEM DESCRIPTION

FREQUENCY STANDARD, CESIUM

The General Services Administration has authorized
the use of this Commercial Item Description (CID) for all federal agencies

1. SCOPE.

This CID describes a Cesium Frequency Standard This CID is meant as a minimum requirement for the Frequency Standard in which only those manufacturers that meet or surpass the following equirements are qualified per this CID

2. CLASSIFICATION.

Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description The equipment shall be Type III, Class 5, Style E and Color R in accordance with MIL-T-28800

3. SALIENT CHARACTERISTICS.

3 1 Design and Construction. The equipment shall be UL (Underwriters Laboratories) listed and approved and/or shall comply with the type III safety requirements of MIL-T-28800

3 2 Leakage Current Protection. Protection shall be provided from leakage current in excess of 3.5 milliamperes (ma) rms (AC and DC) from any accessible conductive parts of equipment (including control shafts with knobs broken or removed and recessed calibration or adjustment controls) to either pole of the power source for any position of the power switch(es)

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to The Resources and Logistics Services Center, SA-ALC/TILDD, Bldg. 171, Post C-12, 485 Quentin Roosevelt Rd., Kelly AFB, TX 78241-6524

AMSC N/A FSC 6625

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3 3 Radio Frequency Interference (RFI) Capacitors Any capacitor used in RFI filters shall be rated for at least four (4) times the operating voltage

3 4 Ventilation Openings

(a) Ventilation openings shall be designed to prevent accidental contact with live electrical circuits and mechanical moving parts

(b) Top ventilation openings shall incorporate small mesh screening or similar design to prevent the expulsion of hot electronic particles due to failure of electronic devices such as capacitors

3 5 Design and Construction. The equipment shall be of solid state construction

3 6 Type of Construction Construction of the Frequency Standard shall be of a monolithic or a separate component design

3 7 Signal Input and Output Connectors Unless otherwise specified all signal input and output connectors shall be type BNC female

3 8 Internal Cooling The equipment shall be capable of operating continuously for twenty-four (24) hours at 50°C If forced air is used it shall be as specified in MIL-T- 28800

3 9 Forced Air Cooling Motor Brushless motors shall be utilized for the forced air cooling motor in the equipment

3 10 Air Filter Equipment utilizing a cooling scheme which exhausts air from the equipment through the cooling fan shall not require an externally accessible air filter

3 11 Thermal Switch A thermal switch shall be provided which will interrupt circuit power to the frequency standard when the internal temperature rises to the degree of possible damage to the electronic components A minimum of 10°C hysteresis drop from this tripping temperature must occur before circuit power is again applied

3 12 Electrical Power Sources The equipment shall operate from nominal commercial, military, and shipboard power sources of 100, 110, 120, 200, 220 and 240 volts ac (+10%) at line frequencies of 50 Hz to 400 Hz, and also shall operate from a rechargeable, internal battery power source for at least 0 5 hours The equipment shall also operate from an external 22 to 28 VDC power source

3 13 Electrical Power Connectors The equipment shall operate with a ground at the primary source on either side of the single phase input

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3 14 Power Consumption Maximum power consumption after warm up of all components combined shall not exceed 85 watts

3 15 Fuses Fuse(s) shall be provided, on the exterior rear panel for replacement without entering the enclosure The fuse(s) shall automatically disconnect the equipment from the power source in the event of excessive current requirements

3 16 Surge and Transient Voltage Protection The Frequency Standard shall contain surge and transient voltage protection internal of the enclosure The transient voltage protection circuit shall be so designed as to prevent electronic device damage within the system per type III solid state tolerance bands, the surge voltage protection circuit shall conform to IEEE 587-1980 The surge protection circuit shall drain excess power due to surges directly to the ground conductor for personnel protection Normal operation of the frequency standard may be inhibited and false operational signals may be generated under input voltage transients greater than ±20 percent and frequency transient conditions greater than ±5 percent from within the nominal type III steady-state tolerance bands damage (other than fuse failure) to the frequency standard shall not occur during voltage transient conditions greater than ± 20 percent and frequency transient conditions greater than ±5 percent within the maximum limits (30 percent voltage transients and 10 percent frequency transients) Normal operation of the frequency standard shall be automatically resumed when the transient has passed and any blown fuses are replaced Blown fuses shall be externally accessible Testing shall be in accordance

3 17 Dimensions and Weight The size and weight of individual components shall be consistent with current commercial capabilities and within the following

(a) The nominal dimensions per component for the Frequency Standard should be approximately 7 inches high, 20 inches wide, and 23 inches deep

(b) The total weight including all accessories (excluding manual) per component, shall not exceed 75 lbs

3 18 Enclosure Requirements Unless otherwise specified herein, the equipment shall comply with the Style E enclosure requirements

3 19 Rack Mount Rack mount capability shall be available as an option for the equipment specified herein

3.20 Reusable Pouch or Container An attached accessory pouch may be used for storage of all supplied accessories in lieu of front-cover storage

3 21 Marking and Identification The front panel marking shall comply with the type III parts, materials and processes panel marking requirements of MIL-T-28800, however, marking identification processes other than the type III requirements specified in MIL-T-28800 may be used, provided the marking and identification processes are capable of passing the front panel marking test when tested in accordance with this CID

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3 22 Environmental Requirements

(a) Temperature Requirements Unless otherwise specified herein the equipment shall meet its specified performance characteristics and other requirements when subjected to the Class 5 temperature requirements and as listed below

(1) Non-operating Temperature The equipment shall meet its performance and accuracy requirements in an non-operating environment of -20°C to -50°C

(2) Operating Temperature The equipment shall meet its performance and accuracy requirements in an operating environment of 0°C to 50°C

(b) Humidity Requirements The equipment shall meet its required performance and accuracy requirements under humidity requirements for Class 5 equipment as specified

(c) Altitude Requirements The equipment shall meet its required performance and accuracy requirements under the altitude requirements for Class 5 equipment as specified

(e) Electromagnetic Interference (EMI) The equipment shall comply with MIL-T-28800

(f) Magnetic Environment The equipment shall comply with the magnetic environment requirements

3 23 Reliability Reliability verification testing, shall be in accordance with the Type III Class 5 environmental conditions as specified in MIL-T-28800

(a) Mean Time Between Failure (MTBF) The equipment's MTBF shall be 2,000 hours The lower test MTBF shall be 2 000 hours and the upper test shall be MTBF 4 000 hours

(b) Pre-conditioning Requirements Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn in and thermal cycling with thermal cycling being accomplished in the sample testing phase

(1) Burn-in Burn-in shall expose the equipment to extended periods of elevated temperatures (40°C for a minimum of 100 hours) Each equipment which fails during burn-in shall be repaired and fully retested until failure free Failures which occur during the burn-in test shall be noted and reported but shall not count toward the establishment of MTBF Prior to and following burn-in, a satisfactory operation test shall be performed A Satisfactory Operation Check (SOC) shall be conducted at selected times during the burn-in test

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(2) Thermal Cycling Thermal Cycling shall expose the equipment to hot and cold cycles for Class 5 equipment per MIL-T-28800 Each equipment which fails during thermal cycling shall be repaired and full cycling test shall be noted and reported but shall not count toward the cycling A Satisfactory Operation Check (SOC) shall be conducted during the last two upper and lower temperature limits

3 24 Calibration and Maintenance Aqjustments The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments These adjustments shall be provided by variable value components which are adjustable by the use of simple means The calibration by substitution of selected components or parts is unacceptable unless specifically approved The calibration adjustments wherever possible, shall be accessible without removal of the instrument case or modules

3 25 Interchangeability The equipment shall meet the Type III interchangeability requirements specified in MIL-T-28800

3 26 Digital Interface The equipment shall meet the digital interface requirements per MIL-T-28800

3 27 Performance Characteristics

(a) Frequency Output The equipment shall provide the following sinusoidal output frequencies through separate connectors on the front and rear panels 100 KHz 1 MHz 5 MHz and 10 MHz This output shall be at least 1 volt rms into a 50 ohm load

(b) Frequency Stability

(1) Accuracy The accuracy shall meet or exceed $1x10^{-11}$ at operating temperature range

(2) Reproducibility The reproducibility shall meet or exceed $5x10^{-12}$

(3) Settings The setting error shall not exceed $7x10^{-13}$

(4) Long Term The long term stability shall meet or exceed $5x10^{-12}$ for the whole tube life

(5) Short Term The short term stability at 5 MHz shall have the following as an absolute minimum acceptable specification

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t (s)	f/f
10^{-1}	5.6×10^{-11}
1	5.6×10^{-11}
10	2.5×10^{-11}
10^{2}	8×10^{-12}
10^{3}	2.5×10^{-12}

(6) Magnetic Field At the 2 gauss DC magnetic field the frequency change shall not exceed 2×10^{-12}

(7) Warm Up The warm up time shall not exceed 45 minutes

(c) Distortion

(1) Harmonic The harmonic distortion shall be down more than 40 dB below rated output

(2) Non-harmonic The non-harmonic distortion shall be down 80 dB or more than rated output

(d) Single Side Band Phase Noise System A signal with 1 Hz bandwidth shall have a signal-to-noise (S/N) ratio with offset from carrier meeting the following as an absolute minimally acceptable level

Hz	dB
10^{-3}	-8
10^{-2}	-28
10^{-1}	-48
1	-82
10	-120
102	-125
103	-140

3.28 Miscellaneous

(a) Internal Battery Charger The Frequency Standard shall be equipped with an internal battery charger, the charging time shall not exceed 16 hours

(b) Power Supply Cords The Frequency Standard shall be equipped with power cords which will allow the unit to be powered from the DC and AC sources

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3 29 Calibration Interval After calibration, the equipment shall meet each performance requirement within the tolerances specified for a period of at least 9 months, based on an operating time of 1500 hours

4 REGULATORY REQUIREMENTS.

In accordance with section 23 403 of the Federal Acquisition Regulations the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials

5 PACKAGING.

The items shall be packed in accordance with normal commercial practice and packed to assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments handling, and storage The shipping shall be in compliance with national motor freight classification and uniform freight classification Shipping container shall be marked as specified in the contract or order

6 QUALITY ASSURANCE.

Certification The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices

The Government reserves the right to require proof of such conformance prior to first delivery, and thereafter, as may be otherwise provided for under the provisions of the contract
Small businesses that cannot offer products with established commercial market acceptability, may compete by submission of bid samples (The Government reserves the rights to require proof of such conformance before first delivery and thereafter as may be otherwise provided for under the provisions of the contract)

7. NOTES.

Metrics Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No 376, and all other requirements of this Commercial Item Description are met

Federal and Military specifications are available from the Defense Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094

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Known Acceptable Products. The following product is known to have previously met the Salient Characteristics of this CID and need not resubmit bid samples or certification All other manufacturers must meet the requirements of this CID and submit to bid testing and certification

Manufacturer	Model	NSN
Hewlett - Packard	5061A-003-E21	6625-00-890-6493
Hewlett - Packard	5061B-003-E21	6625-01-318-5654

MILITARY INTERESTS

Custodian.
Air Force - 99

Reviewing Activites.
Army - ER
Navy - AS
DLA - ES

PREPARING ACTIVITY

Air Force - 82

Project Nos 6625-0736
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